Chapter 3. Convergence of Fourier Series

Mean-square convergence and pointwise behaviour

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3.1 Mean-square convergence of Fourier series

Motivation. $S_N(f)$ is the orthogonal projection onto trigonometric polynomials of degree $\le N$, which are dense, so every integrable f is the mean-square limit of its Fourier series.

Theorem 1.1 (Mean-square convergence)

If f is integrable on the circle, then

$$\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2\, d\theta \to 0 \quad \text{as } N \to \infty.$$

3.1.1 Vector spaces and inner products

An **inner product** on a complex vector space V is linear in the first variable, $(Y, X) = \overline{(X, Y)}$, $(X, X) \ge 0$; **strictly positive definite** if $(X, X) = 0$ only for $X = 0$. $\|X\| = (X, X)^{1/2}$. For any inner product:

- **Pythagoras:** if $X \perp Y$ ($(X, Y) = 0$), then $\|X + Y\|^2 = \|X\|^2 + \|Y\|^2$.
- **Cauchy–Schwarz:** $|(X, Y)| \le \|X\|\|Y\|$.
- **Triangle inequality:** $\|X + Y\| \le \|X\| + \|Y\|$.

Proofs [Supplement]. Pythagoras: expand. Cauchy–Schwarz: if $\|Y\| = 0$, $0 \le \|X - tY\|^2 = \|X\|^2 - 2t\,\mathrm{Re}(X, Y)$ for all real t forces $\mathrm{Re}(X, Y) = 0$, likewise $\mathrm{Im}(X, Y) = 0$. Otherwise $c = (X, Y)/\|Y\|^2$ gives $X - cY \perp Y$ and $\|X\|^2 = \|X - cY\|^2 + |c|^2\|Y\|^2 \ge |(X, Y)|^2/\|Y\|^2$. Triangle: $\|X + Y\|^2 = \|X\|^2 + 2\,\mathrm{Re}(X, Y) + \|Y\|^2 \le (\|X\| + \|Y\|)^2$.

Examples. (1) $\mathbb{R}^d$, $\mathbb{C}^d$ with the usual inner products.

(2) $\ell^2(\mathbb{Z})$: sequences $(a_n)_{n\in\mathbb{Z}}$ with $\sum |a_n|^2 < \infty$, $(A, B) = \sum a_n \bar{b}_n$ (finite by Cauchy–Schwarz). A vector space ($|a + b|^2 \le 2|a|^2 + 2|b|^2$), and **complete**.

(3) $\mathcal{R}$: integrable functions on $[0, 2\pi]$, $(f, g) = \frac{1}{2\pi}\int_0^{2\pi} f\bar{g}$. Not strictly positive definite ($f = 0$ off one point has $\|f\| = 0$) and not complete; its completion is Lebesgue's L^2.

3.1.2 Proof of mean-square convergence

$e_n(\theta) = e^{in\theta}$ is orthonormal in $\mathcal{R}$: $(e_n, e_m) = \frac{1}{2\pi}\int e^{i(n-m)\theta} d\theta = \delta_{nm}$. With $a_n = \hat{f}(n) = (f, e_n)$, $S_N(f) = \sum_{|n|\le N} a_n e_n$.

Key orthogonality. $(f - S_N(f), e_n) = a_n - a_n = 0$ for $|n| \le N$, so $f - S_N(f) \perp \sum_{|n|\le N} b_n e_n$, and by Pythagoras

$$\|f\|^2 = \|f - S_N(f)\|^2 + \sum_{|n|\le N} |a_n|^2. \tag{3.1}$$

Lemma 1.2 (Best approximation)

If f is integrable on the circle with Fourier coefficients a_n, then $\|f - S_N(f)\| \le \|f - \sum_{|n|\le N} c_n e_n\|$ for all complex c_n, with equality iff $c_n = a_n$ for all $|n| \le N$.

Proof. $f - \sum c_n e_n = (f - S_N(f)) + \sum b_n e_n$, $b_n = a_n - c_n$, orthogonal summands, so $\|f - \sum c_n e_n\|^2 = \|f - S_N(f)\|^2 + \sum |b_n|^2$. ∎

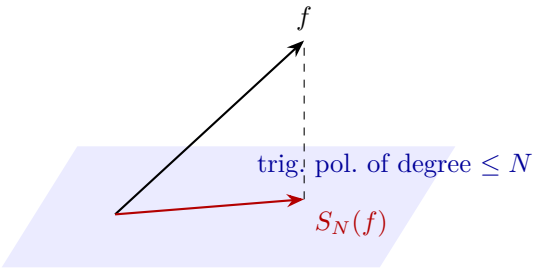


Figure. *$S_N(f)$ is the orthogonal projection of f, the best approximation (Lemma 1.2).*

Proof of Theorem 1.1.

Step 1 (continuous f). Corollary 5.4 gives a trigonometric polynomial P of degree M with $|f-P| < \epsilon$, so $\|f-P\| < \epsilon$, and by Lemma 1.2 $\|f - S_N(f)\| < \epsilon$ for $N \ge M$.

Step 2 (integrable f). Let $|f| \le B$. Lemma 3.2 gives continuous g, $|g| \le B$, $\int |f-g| < \epsilon^2$, so $\|f-g\|^2 \le \frac{2B}{2\pi}\int |f-g| < C\epsilon^2$. With $\|g-P\| < \epsilon$ (Step 1), $\|f-P\| < C'\epsilon$, and Lemma 1.2 gives $\|f - S_N(f)\| < C'\epsilon$ for large N. ∎

Theorem 1.3 (Parseval)

Let $f \sim \sum a_n e^{in\theta}$ be integrable on the circle. Then:
(i) Mean-square convergence: $\frac{1}{2\pi}\int_0^{2\pi} |f(\theta) - S_N(f)(\theta)|^2 d\theta \to 0$.
(ii) Parseval's identity: $\sum_{n=-\infty}^{\infty} |a_n|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f(\theta)|^2 d\theta$.
(iii) The map $f \mapsto (a_n)_{n\in\mathbb{Z}}$ is "unitary": $(f,g) = \sum a_n \bar{b}_n$ for $g \sim \sum b_n e^{in\theta}$ (Lemma 1.5).

Proof. (i) is Theorem 1.1.

(ii) Let $N \to \infty$ in (3.1).

(iii) is Lemma 1.5 below. ∎

Bessel's inequality $\sum |a_n|^2 \le \|f\|^2$ follows from (3.1).

Theorem 1.4 (Riemann–Lebesgue lemma)

If f is integrable on the circle, then $\hat{f}(n) \to 0$ as $|n| \to \infty$. Equivalently, $\int_0^{2\pi} f(\theta)\sin N\theta\, d\theta \to 0$ and $\int_0^{2\pi} f(\theta)\cos N\theta\, d\theta \to 0$ as $N \to \infty$.

Proof. Parseval gives $\sum |\hat{f}(n)|^2 < \infty$; for the reformulation write $\cos N\theta$, $\sin N\theta$ via $e^{\pm iN\theta}$. ∎

Lemma 1.5

If $F \sim \sum a_n e^{in\theta}$ and $G \sim \sum b_n e^{in\theta}$ are integrable on the circle, then $\frac{1}{2\pi}\int_0^{2\pi} F(\theta)\overline{G(\theta)}\, d\theta = \sum_{n=-\infty}^{\infty} a_n \bar{b}_n$, the series converging absolutely.

Proof. [Supplement] Absolute convergence: $|a_n\bar{b}_n| \le \frac{1}{2}(|a_n|^2 + |b_n|^2)$ and Parseval. Polarization: $(F,G) = \frac{1}{4}\sum_{k=0}^{3} i^k \|F + i^k G\|^2$ and likewise $\sum a_n\bar{b}_n = \frac{1}{4}\sum_k i^k \sum_n |a_n + i^k b_n|^2$; $F + i^k G$ has coefficients $a_n + i^k b_n$, so Parseval applies termwise. ∎

3.2 Return to pointwise convergence

Motivation. Differentiability at a point gives convergence there, and convergence is local; continuity alone does not suffice.

3.2.1 A local result

Theorem 2.1

If f is integrable on the circle and differentiable at θ_0, then $S_N(f)(\theta_0) \to f(\theta_0)$.

Proof. Define, for $|t| \le \pi$,

$$F(t) = \begin{cases} \dfrac{f(\theta_0 - t) - f(\theta_0)}{t} & t \neq 0, \\ -f'(\theta_0) & t = 0. \end{cases}$$

F is bounded near 0 and integrable on $[-\pi, \pi]$ (Proposition 1.4). As $\frac{1}{2\pi}\int D_N = 1$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int_{-\pi}^{\pi} (f(\theta_0 - t) - f(\theta_0)) D_N(t)\, dt = \frac{1}{2\pi}\int_{-\pi}^{\pi} F(t)\, t\, D_N(t)\, dt.$$

With $\sin((N + \frac{1}{2})t) = \sin Nt \cos\frac{t}{2} + \cos Nt \sin\frac{t}{2}$,

$$S_N(f)(\theta_0) - f(\theta_0) = \frac{1}{2\pi}\int F(t)\frac{t}{\sin(t/2)} \cos\tfrac{t}{2}\, \sin Nt\, dt + \frac{1}{2\pi}\int F(t)\, t \cos Nt\, dt.$$

$F(t)\frac{t\cos(t/2)}{\sin(t/2)}$ and $tF(t)$ are integrable ($t/\sin(t/2)$ is bounded), so both integrals tend to 0 by Theorem 1.4. ∎

(The proof only needs $|f(\theta) - f(\theta_0)| \le M|\theta - \theta_0|$.)

Theorem 2.2 (Riemann localization principle)

If f, g are integrable on the circle and $f = g$ on an open interval I containing θ_0, then $S_N(f)(\theta_0) - S_N(g)(\theta_0) \to 0$.

Proof. $f - g$ vanishes on I, so it is differentiable at θ_0 with value 0; apply Theorem 2.1. ∎

3.2.2 A continuous function with diverging Fourier series

Symmetry breaking. The odd sawtooth $f(\theta) = i(\pi - \theta)$ on $0 < \theta < \pi$ has $f \sim \sum_{n\neq 0} \frac{e^{in\theta}}{n}$. The half $\sum_{n\le -1} \frac{e^{in\theta}}{n}$ is the Fourier series of no integrable $\tilde f$: else $|A_r(\tilde f)(0)| = \sum_{n\ge 1} \frac{r^n}{n} \to \infty$, while $|A_r(\tilde f)(0)| \le \sup|\tilde f|$. For $N \ge 1$ let

$$f_N(\theta) = \sum_{1\le|n|\le N} \frac{e^{in\theta}}{n}, \qquad \tilde f_N(\theta) = \sum_{-N\le n\le -1} \frac{e^{in\theta}}{n}.$$

Then (i) $|\tilde f_N(0)| = \sum_{n=1}^{N} \frac{1}{n} \ge \log N$, and (ii) $f_N(\theta)$ is bounded uniformly in N, θ by the lemma below.

Lemma 2.3

Suppose the Abel means $A_r = \sum_{n\ge 1} r^n c_n$ of $\sum c_n$ are bounded as $r \to 1^-$, and $c_n = O(1/n)$. Then the partial sums $S_N = \sum_{n=1}^{N} c_n$ are bounded.

Proof. Let $r = 1 - 1/N$, $n|c_n| \le M$. Then

$$|S_N - A_r| \le \sum_{n=1}^{N} |c_n|(1-r^n) + \sum_{n>N} r^n|c_n| \le M\sum_{n=1}^{N}(1-r) + \frac{M}{N}\sum_{n>N} r^n \le MN(1-r) + \frac{M}{N}\frac{1}{1-r} = 2M,$$

as $1-r^n \le n(1-r)$. So $|A_r| \le M$ gives $|S_N| \le 3M$. ■

Apply it with $c_n = \frac{e^{in\theta}}{n} - \frac{e^{-in\theta}}{n}$ (partial sums $f_N(\theta)$, $|c_n| \le 2/n$): the Abel means $(f * P_r)(\theta)$ are bounded by $\sup|f|$. So $|f_N(\theta)| \le C$.

Shift frequencies by $2N$: $P_N = e^{i2N\theta} f_N$, $\tilde{P}_N = e^{i2N\theta}\tilde{f}_N$, with frequencies in $N \le n \le 3N$, $n \ne 2N$ (resp. $N \le n \le 2N-1$).

Lemma 2.4

$S_M(P_N) = P_N$ if $M \ge 3N$; $S_M(P_N) = \tilde{P}_N$ if $M = 2N$; $S_M(P_N) = 0$ if $M < N$.

Proof. S_M keeps the frequencies $|n| \le M$; those of P_N lie in $[N, 3N]$, and those $\le 2N$ are the frequencies of $\tilde{P}_N$. ■

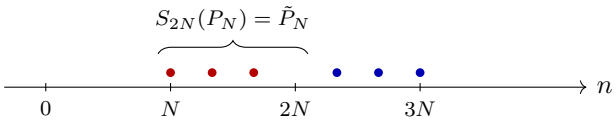


Figure. *Frequencies of P_N; S_{2N} keeps the left half.*

The construction. Take $\alpha_k > 0$, $\sum \alpha_k < \infty$, and $N_{k+1} > 3N_k$ with $\alpha_k \log N_k \to \infty$; e.g. $\alpha_k = 1/k^2$, $N_k = 3^{2^k}$ ($\alpha_k \log N_k = \frac{2^k \log 3}{k^2} \to \infty$). Let

$$f(\theta) = \sum_{k=1}^{\infty} \alpha_k P_{N_k}(\theta).$$

As $|P_N| = |f_N| \le C$, the series converges uniformly: f is continuous with termwise coefficients. By Lemma 2.4 ($2N_m < N_k$ for $k > m$, $2N_m \ge 3N_k$ for $k < m$),

$$S_{2N_m}(f) = \sum_{k<m} \alpha_k P_{N_k} + \alpha_m \tilde{P}_{N_m}.$$

At 0: $|\sum_{k<m} \alpha_k P_{N_k}(0)| \le C \sum \alpha_k$ and $|\tilde{P}_{N_m}(0)| \ge \log N_m$, so $|S_{2N_m}(f)(0)| \ge \alpha_m \log N_m - C' \to \infty$: *the Fourier series of the continuous f diverges at 0.*

3.3 Selected exercises

Motivation. Consequences of Parseval and Riemann–Lebesgue: Wirtinger–Poincaré inequalities, the Dirichlet integral, and absolute convergence for C^1 and Lipschitz f.

Exercise 3.11 (Wirtinger and Poincaré inequalities)

(a) Let f be T-periodic, continuous and piecewise C^1 with $\int_0^T f = 0$. Show that $\int_0^T |f|^2 \le \frac{T^2}{4\pi^2} \int_0^T |f'|^2$, with equality iff $f(t) = A\sin(2\pi t/T) + B\cos(2\pi t/T)$.
(b) Let f be C^1 on $[a,b]$ with $f(a) = f(b) = 0$. Show that $\int_a^b |f|^2 \le \frac{(b-a)^2}{\pi^2} \int_a^b |f'|^2$, with equality iff $f(t) = A\sin(\pi \frac{t-a}{b-a})$; in particular the constant is optimal.

Solution.

(a)

Step 1 (reduction to $T = 2\pi$). $g(s) = f(Ts/2\pi)$ has $\int_0^{2\pi} |g|^2 = \frac{2\pi}{T}\int_0^T |f|^2$ and $\int_0^{2\pi} |g'|^2 = \frac{T}{2\pi}\int_0^T |f'|^2$, so the claim becomes $\int_0^{2\pi} |g|^2 \le \int_0^{2\pi} |g'|^2$, with corresponding extremals.

Step 2 (coefficients of g'). For $T = 2\pi$, integrating by parts on each C^1 piece (boundary terms telescope and cancel by periodicity), $\hat{f'}(n) = in\hat{f}(n)$; and $\hat{f}(0) = 0$.

Step 3. Parseval for f and f':

$$\frac{1}{2\pi}\int_0^{2\pi} |f|^2 = \sum_{n\neq 0} |\hat{f}(n)|^2 \le \sum_{n\neq 0} n^2 |\hat{f}(n)|^2 = \frac{1}{2\pi}\int_0^{2\pi} |f'|^2.$$

Step 4 (equality). Iff $(n^2 - 1)|\hat{f}(n)|^2 = 0$ for all n, i.e. $\hat{f}(n) = 0$ for $|n| \ge 2$; then $f = \hat{f}(-1)e^{-it} + \hat{f}(1)e^{it} = A\sin t + B\cos t$ by Corollary 2.2. Conversely these give equality.

(b)

Step 1. Take $a = 0$, $L = b$. Extend f oddly to $[-L, L]$ and $2L$-periodically; $f(0) = f(L) = 0$ makes $\tilde{f}$ continuous and piecewise C^1, with $\int_{-L}^{L} \tilde{f} = 0$.

Step 2. By (a) with $T = 2L$, $\int_{-L}^{L} |\tilde{f}|^2 \le \frac{L^2}{\pi^2}\int_{-L}^{L} |\tilde{f}'|^2$, and each side is twice its integral over $[0, L]$.

Step 3. Equality forces $\tilde{f} = A\sin(\pi t/L) + B\cos(\pi t/L)$, and oddness $B = 0$; $\sin(\pi t/L)$ attains it, so L^2/π^2 is optimal. ■

Exercise 3.12 (The Dirichlet integral)

Prove that $\int_0^\infty \frac{\sin x}{x}\,dx = \frac{\pi}{2}$ (as an improper Riemann integral).

Solution.

Step 1 (existence). For $R > 1$, by parts $\int_1^R \frac{\sin x}{x}\,dx = \cos 1 - \frac{\cos R}{R} - \int_1^R \frac{\cos x}{x^2}\,dx$, convergent as $R \to \infty$; the integrand is continuous on $(0, 1]$ with limit 1 at 0. So $I = \lim_R \int_0^R \frac{\sin x}{x}\,dx$ exists; take $R_N = (N + \frac{1}{2})\pi$.

Step 2 (splitting D_N). $\frac{1}{2\pi}\int_{-\pi}^{\pi} D_N = 1$. Let $h(\theta) = \frac{1}{\sin(\theta/2)} - \frac{2}{\theta}$ for $0 < |\theta| \le \pi$, $h(0) = 0$. As

$$h(\theta) = \frac{\theta - 2\sin(\theta/2)}{\theta\sin(\theta/2)} = \frac{\theta^3/24 + O(\theta^5)}{\theta^2/2 + O(\theta^4)} \to 0,$$

h is continuous on $[-\pi, \pi]$.

Step 3 (Riemann–Lebesgue). Expanding $\sin((N + \frac{1}{2})\theta)$ as in Theorem 2.1, Theorem 1.4 gives $\int_{-\pi}^{\pi} h(\theta)\sin((N + \frac{1}{2})\theta)\,d\theta \to 0$, so

$$\int_{-\pi}^{\pi} \frac{2\sin((N + \frac{1}{2})\theta)}{\theta}\,d\theta = 2\pi - \int_{-\pi}^{\pi} h(\theta)\sin\left((N + \tfrac{1}{2})\theta\right)d\theta \to 2\pi.$$

Step 4. By evenness $\int_0^{\pi} \frac{\sin((N+\frac{1}{2})\theta)}{\theta}d\theta \to \frac{\pi}{2}$, which is $\int_0^{R_N} \frac{\sin x}{x}\,dx$ after $x = (N + \frac{1}{2})\theta$. So $I = \frac{\pi}{2}$. ■

Exercise 3.14 (C^1 functions have absolutely convergent Fourier series)

Prove that the Fourier series of a C^1 function f on the circle converges absolutely, hence uniformly to f.

Solution.

Step 1. As in Exercise 3.11, $\hat{f'}(n) = in\hat{f}(n)$, so $\sum_n n^2|\hat{f}(n)|^2 = \frac{1}{2\pi}\int |f'|^2 < \infty$.

Step 2. By Cauchy–Schwarz,

$$\sum_{0<|n|\le N} |\hat f(n)| = \sum_{0<|n|\le N} |n\hat f(n)|\cdot\frac{1}{|n|} \le \Big(\sum_{n} n^2|\hat f(n)|^2\Big)^{1/2}\Big(\sum_{n\neq 0}\frac{1}{n^2}\Big)^{1/2}.$$

uniformly in N, so $\sum|\hat f(n)|<\infty$.

Step 3. Corollary 2.3 gives uniform convergence, improving Corollary 2.4 (C^2). ∎

Exercise 3.16 (Lipschitz functions: Bernstein's argument)

Let f be 2π-periodic with $|f(x)-f(y)|\le K|x-y|$. Prove that $\sum_n|\hat f(n)|<\infty$, so the Fourier series of f converges absolutely and uniformly to f.

Solution.

Step 1. For $h>0$, $g_h(x)=f(x+h)-f(x-h)$ has $\hat g_h(n)=2i\sin(nh)\hat f(n)$. Parseval and $|g_h|\le 2Kh$ give

$$\sum_n 4\sin^2(nh)|\hat f(n)|^2 = \frac{1}{2\pi}\int_0^{2\pi}|g_h|^2 \le 4K^2h^2, \qquad \text{so} \qquad \sum_n \sin^2(nh)|\hat f(n)|^2 \le K^2h^2.$$

Step 2. For $p\ge 0$, $h=\pi/2^{p+1}$ and $2^{p-1}<|n|\le 2^p$: $\frac{\pi}{4}<|n|h\le\frac{\pi}{2}$, so $\sin^2(nh)\ge\frac12$ and

$$\sum_{2^{p-1}<|n|\le 2^p}|\hat f(n)|^2 \le 2K^2h^2 = \frac{K^2\pi^2}{2^{2p+1}}.$$

Step 3. The block has at most 2^{p+1} integers, so by Cauchy–Schwarz

$$\sum_{2^{p-1}<|n|\le 2^p}|\hat f(n)| \le (2^{p+1})^{1/2}\Big(\frac{K^2\pi^2}{2^{2p+1}}\Big)^{1/2} = \frac{K\pi}{2^{p/2}}.$$

Step 4. The blocks cover $n\neq 0$ and $\sum_p 2^{-p/2}<\infty$, so $\sum_n|\hat f(n)|<\infty$; Corollary 2.3 gives uniform convergence. (Likewise for Hölder exponent $\alpha>\frac12$.) ∎